Quantification: 543_3
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	13
interior lacuna count	12
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 12)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2350.2	2188.5	668.925	px^2
major_axis_length_px	lacuna shape	91.5632	84.5141	33.0603	px
minor_axis_length_px	lacuna shape	36.5948	35.0553	10.2044	px
aspect_ratio	lacuna shape	2.6601	2.3867	1.276	unitless
eccentricity	lacuna shape	0.889	0.9075	0.078	unitless
solidity	lacuna shape	0.8742	0.9204	0.1013	unitless
roots_count	headline	7	5	4.671	count
ring_length_r30_px	headline	325.583	358	143.164	px
ring_length_r60_px	reported	941.5	1003	285.113	px
owned_length_px	ownership-dependent	3319.04	2948.86	1756.14	px
edge_count	ownership-dependent	153.083	129	84.1043	count
mean_edge_length_px	ownership-dependent	22.1358	21.6998	2.0458	px
perimeter_px	normalised	221.015	219.693	100.917	px
ring_area_r30_px2	normalised	8668.5	8576.5	2568.76	px^2
ring_area_r60_px2	normalised	21815.7	21875	4394.01	px^2
in_frame_fraction_r30	normalised	0.9974	1	0.009	unitless
in_frame_fraction_r60	normalised	0.9801	1	0.058	unitless
ring_density_r30	normalised	0.0366774	0.0364641	0.0101872	px^-1
ring_density_r60	normalised	0.0431053	0.0442457	0.00925126	px^-1
roots_per_100px_perimeter	normalised	3.1711	3.5359	1.1133	per 100 px
ring_attached_length_r30_px	network v2	288.667	332.5	149.507	px
ring_attached_length_r60_px	network v2	752.25	909.5	370.416	px
ring_length_w_r30_px	network v2	369.153	404.791	163.984	px
ring_length_w_r60_px	network v2	1072.97	1142.5	323.917	px
sholl_crossings_r10	network v2	8.8333	9	4.6677	count
sholl_crossings_r20	network v2	11.0833	12.5	4.1442	count
sholl_crossings_r30	network v2	13.5	13.5	4.5025	count
loop_count	loops, report only	17.3333	14	11.9418	count
loop_edge_share	loops, report only	0.4035	0.4063	0.109	share
ring_width_mean_r30_px	width	7.4443	7.6762	0.9107	px

Per field
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measure	value	units
analysed_area_px2	1.02406e+06	px^2
total_skeleton_length_px	41269	px
canalicular_length_density_per_px	0.0402993	px^-1
junction_count	1070	count
junction_density_per_px2	0.00104486	px^-2
lacuna_count	13	count

lacunae_per_px2	1.269e-05	px^-2
skeleton_component_count	149	count
mean_component_length_px	276.973	px
median_component_length_px	19	px
field_length_density_w_per_px	0.0459196	px^-1
field_density_without_flagged_per_px	0.0402993	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	224	px
unattached_component_count	15	count
canalicular_length_density_attached_per_px	0.0402993	px^-1
field_length_density_with_unattached_per_px	0.040518	px^-1
loop_count	244	count
loop_edge_share	0.4439	share
edge_count	2138	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.9145	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10.198	px
width_pixels_used	41269	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2174	80.8669	35.2412	2.2947	0.9	0.9518	0.4109	79.3965	270.898	False
2	3447	90.3677	63.9486	1.4131	0.7066	0.674	0.4223	197.059	916.902	False
3	1883	69.2617	34.8694	1.9863	0.864	0.9448	0.5742	250.833	340.47	False
4	3536	181.867	30.1414	6.0338	0.9862	0.7148	0.0738	327.046	42.4075	False
5	2052	69.2015	39.4228	1.7554	0.8219	0.9112	0.0622	388.671	369.939	False
6	2266	88.1613	35.5691	2.4786	0.915	0.9021	0.1324	473.05	745.191	False
7	1625	80.1248	27.6985	2.8927	0.9383	0.8325	0.2321	545.572	877.076	False
8	1683	71.0123	31.3735	2.2634	0.8971	0.9355	-0.0518	634.68	198.315	False
9	2203	91.2092	31.4053	2.9043	0.9389	0.9295	0.0832	734.301	635.44	False
10	2633	93.5592	36.2778	2.579	0.9218	0.9454	0.0457	780.576	481.495	False
11	531	28.7881	24.8252	1.1596	0.5063	0.9332	0.5942	1012.28	184.834	True
12								458.12	481.03	False
13								857.425	285.616	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	7	336	1050	1380.05	63	21.9056	7.4363
2	15	572	1426	4508.26	221	20.3994	7.2258
3	8	380	1030	5869.85	275	21.3449	8.3846
4	16	458	976	5652.9	263	21.4939	8.1099
5	5	225	726	1034.78	40	25.8695	7.2282
6	11	406	1130	3234.03	146	22.1509	8.1001

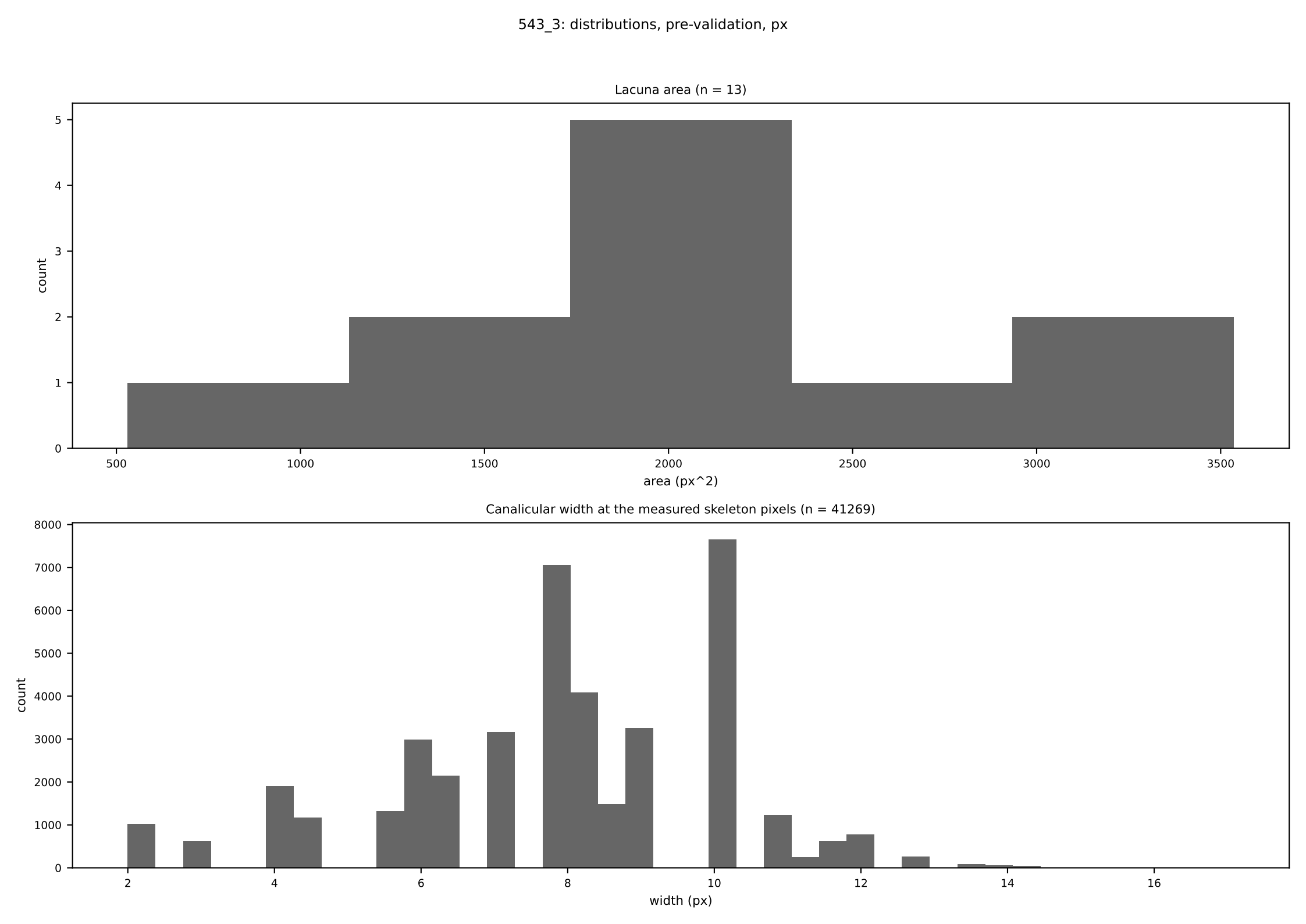
7	5	426	1281	4505.04	226	19.9338	6.8746
8	3	134	527	1898.07	93	20.4093	7.9116
9	5	409	1115	5254.9	227	23.1493	7.7657
10	4	289	820	2008.44	79	25.4234	7.8079
11	2	136	454	872.009	42	20.7621	6.1456
12	2	119	570	2663.68	112	23.7829	4.9003
13	3	153	647	1818.53	92	19.7667	7.5868

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	210.066	8595	21715	1	0.9616	0.0390925	0.0483537	3.3323
2	363.446	12097	29071	1	1	0.0472844	0.0490523	4.1272
3	184.794	7941	19515	1	1	0.0478529	0.0527799	4.3291
4	415.99	13536	26753	0.9687	0.7992	0.0338357	0.0364819	3.8462
5	205.965	8333	20013	1	1	0.0270011	0.0362764	2.4276
6	241.907	9380	23807	1	1	0.0432836	0.047465	4.5472
7	229.321	8558	22035	1	1	0.049778	0.0581348	2.1804
8	191.965	7943	21525	1	1	0.0168702	0.0244832	1.5628
9	238.108	9212	24077	1	1	0.0443986	0.0463098	2.0999
10	242.936	9386	24420	1	1	0.0307905	0.033579	1.6465
11	95.3909	3354	9525	0.6102	0.5725	0.0405486	0.047664	2.0966
12	47.4558	4187	13513	1	1	0.0284213	0.0421816	4.2144
13	80.2254	4854	15344	1	1	0.0315204	0.0421663	3.7395

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	307	905	385.433	1221.61	8	13	17
2	528	1231	659.156	1624.61	18	16	19
3	358	1014	424.149	1175.66	10	12	18
4	449	914	514.889	1109.35	14	14	9
5	185	518	250.309	831.553	6	9	11
6	394	975	452.877	1279.05	10	13	14
7	371	1180	490.617	1457.24	11	15	20
8	96	272	147.569	589.446	3	5	7
9	376	979	462.191	1254.86	12	14	16
10	222	600	331.321	931.98	8	12	13
11	119	302	152.225	514.546	4	4	6
12	79	190	134.054	663.541	3	3	8
13	99	249	177.267	736.713	3	7	10


543_3: distributions, pre-validation, px
Lacuna area (n = 13)
count
area (px^2)
0
1
2
3
4
5
500
1000
1500
2000
2500
3000
3500
Canalicular width at the measured skeleton pixels (n = 41269)
count
width (px)
0
1000
2000
3000
4000
5000
6000
7000
8000
2
4
6
8
10
12
14
16